Daniel Jang

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Technical Skills

Languages: Java, C, Python, JavaScript, HTML/CSS, TypeScript, SQL, Ruby, Dart
Developer Tools: Git, VS Code, Linux, Docker, GCP, Bash, Maven, Powershell
Frameworks: React, Node.js, Spring Boot, Flutter, Rails, Flask, Vite, Gemini API

Projects

Siel – Digital Biographer | *Java 17, Spring Boot, Node.js, Powershell, Gemini API* — May 2026

- Designed a photo-driven journaling platform to encourage self-reflection through modern technologies, using in-memory processing to extract metadata without committing sensitive media to disk, eliminating the local data footprint.
- Developed a high-concurrency metadata pipeline in **Java 17** using ParallelStreams to execute multi-threaded EXIF extraction, significantly reducing processing time for bulk image batches.
- Hardened system security via Argon2id hashing and RS256 asymmetric **JSON Web Token** signing, integrating the Gemini API to generate journal narratives within a secure environment.

Mappin' – Destination Alarm System | *TypeScript, JavaScript, HTML, CSS* — Apr. 2026

- Constructed a cross-platform location alarm to notify users upon arrival at their intended destinations by implementing Service Worker logic between a Chrome Extension (MV3) and a **Progressive Web app**.
- Optimized data extraction by implementing a URL parser that uses URL APIs to resolve shortened redirects, ensuring successful destination mapping across **30+ varying coordinates** with a **0% failure rate** for common share types.
- Implemented a high-precision arrival detection system by integrating the Haversine formula, allowing the model to calculate real-time optimal distance and trigger synchronized haptic alerts across user-defined radii from **50m** to **2000m**.

UCEY – Urban Planning Interface | *TypeScript, React, Python, Web APIs* — Mar. 2026

- Developed a web app that helps urban planners find underused city land like brownfields and estimate the redevelopment cost using **TypeScript** and **React**.
- Built the full stack design, connecting map APIs and figures from **Statistics Canada** to the interactive interface, so users can browse and assess sites directly on a map rather than manually inspecting raw data.
- Strengthened platform security by including an authentication system featuring **Multi-Factor Authentication** and secure tokens using APIs to ensure protected, personalized user sessions validated by a successful pilot test with **5+** users.

YSafe – Pedestrian Safety Navigation App | *Dart, Google Maps API, ArcGIS, Hive* — Feb. 2026

- Designed a navigation web app that recommends routes based on localized safety scores atop simple distance optimization by integrating **Flutter** with the Google APIs, implementing a custom weighting logic.
- Optimized route-generation by extracting **5 years of York Region Open Data**, allowing the app to recommend routes based on historically occurred safety incidents.
- Improved performance using Riverpod framework and Hive caching, enabling an alert engine to deliver real-time navigation warnings.

Experience & Activities

Waterloo Rocketry — Nov. 2025 – Present
Software Member — *Waterloo, ON*

- Analyzed data collection workflows in **Python** and **Java** to ensure software stability during rocket launch operations.
- Collaborated with the team to develop a full-stack, ground control software using **TypeScript** and **Python**.
- Handled feature stabilization for the July architecture lock to focus on integration testing and debugging to ensure software stability for the upcoming launch.

Waterloo Aerial Robotics Group — Nov. 2025 – Apr 2026
Autonomy Member — *Waterloo, ON*

- Tested autonomous flight performance by engineering a data pipeline for object detection and using the flight-test datasets to analyze the model training accuracy.
- Bridged the gap between the camera and ground control by developing a real-time data stream that processes detected targets, enabling the team to verify autonomous tasks.

Education

University of Waterloo — Waterloo, ON
Bachelor of Computer Science, Honours | Term Distinction — *Sep. 2025 – Apr. 2030*

- **Relevant Coursework:** Algorithms, Memory Management, Data Structures, Imperative & Functional Programming, Shell Scripting, Linear Algebra I, Calculus II, Probability